Helix Forge Research Failure Analysis Research Brief

HELIX FORGE LAB / p04-full-006298 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded failure analysis evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000252 (notebooks/exports/2026-07-13/helix-forge-research/benchmark-design/specifications/helix-forge-research-benchmark-design-slices-v01.js

Slice	Records	Precision	Recall	F1
slice-01	238	0.895	0.847	0.870
slice-02	236	0.877	0.860	0.868
slice-03	206	0.894	0.873	0.883
slice-04	209	0.876	0.849	0.862
slice-05	229	0.893	0.862	0.877